be grown successfully, but tests showed that they could not compete with forage sorghum in either yield or forage quality.

During the drought of 1932-1936, survival of native grasses was studied carefully under the direction of Mr. David A. Savage. In anticipation of a need to reseed after the drought, studies were initiated to determine optimum seeding date, seedbed preparation, and seeding methods. As will be seen, these studies were expanded in later years. The breeding program, or perhaps more accurately a selection program, for improving buffalograss began in 1936. A selection that was superior in seed production and height of seed stems was located and increased vegetatively in 1942. This female plant selection then was pollinated with numerous selected male plants. This improved selection was released as a buffalograss variety named Hays 1-I, and several thousand pounds of seed were harvested and distributed. Concurrent with this buffalograss variety improvement work was the research needed to develop a system for improving the germination of newly harvested buffalograss seed. This process is discussed in some detail in the section on the L.C. Aicher administration. Most of the grass research work in the forage project was terminated with the arrival of Dr. John Launchbaugh in 1955 as leader of the Range Management project. As noted in the Range Management section, he did outstanding work in range reseeding and other aspects of grass management.

It was apparent as early as 1902 that alfalfa could be grown successfully on the bottomland fields along Big Creek. Problems associated with establishment and seed production were studied. Later, lines and varieties, including the bacterial wilt-resistant variety Buffalo, were evaluated. An active alfalfa breeding project was initiated in 1954 when the spotted alfalfa aphid first appeared. The plant breeder, Harold Hackerott, and the entomologist, Tom Harvey, worked together to find resistance to this insect pest, and, certainly in nearly record time, developed and released the resistant variety Cody. A similar program a few years later resulted in the release of Kanza, the first alfalfa variety resistant to the pea aphid. This cooperation between scientists from different disciplines illustrates the value of having close association and cooperation among the resident staff members.

Hackerott transferred to the sorghum breeding position in 1969, and only limited work was accomplished on cultivated forages other than sorghum after that. As noted above, the Range Management scientist did most of the grass research. A new project was started in 1971 to investigate some possible alternative crops for use in west central Kansas.

ALTERNATIVE CROPS

Hard red winter wheat became the crop of choice soon after it was introduced into the area late in the 19th century. However, farmers then and continuing to the present have sought possible alternative crops that would equal or exceed returns from wheat. The first of these crops that became important, of course, was sorghum; but during the early years of the 20th century, considerable acreage of corn also was planted. Since the Station's origin, many other possible alternative crops have been planted and observed by crop scientists. As noted previously, most of these crops were discarded quickly as unsatisfactory for the climatic conditions or simply did not have potential for economic returns that could compete with those of wheat and sorghum. This was true even with corn, especially following development of combine-type grain sorghums. More recently, with improved corn hybrids and soybean varieties, these two crops have extended their area of adaptability further west in Kansas than previously thought possible.

Research to find alternative crops intensified with the arrival of Mr. William D. Stegmeier in 1971. He had worked previously at the Garden City Branch Experiment Station, where his studies with wheat, sorghum, corn, and several possible alternative minor crops

such as castor beans, soybeans, pinto beans, safflower, winter barley, and popcorn provided an excellent background for his studies at Hays. When he came to the Station, most of the oil-type sunflower production was in Texas and in the northern states. Stegmeier believed that the crop had potential in Kansas and began an intensive breeding program to improve yields and increase disease and insect resistance in sunflower varieties. His work contributed to acceptance of the crop by Kansas farmers. Acreage was limited for several years, partly because seed-processing plants were not available within the state. With the addition of such plants, sunflower is now an established crop in the state.

Stegmeier's most notable work was his breeding program with pearl millet. This crop, which is entirely different from the previously tested foxtail and proso millets, is a staple food and feed crop in many countries of Asia and Africa, but it had never been grown in Kansas. He believed that it had potential as a new crop for Kansas farmers and that it might compete successfully with grain sorghum. When he started his work, the typical pearl millet variety was a tall late-maturing type adapted to tropical climates. Stegmeier knew that, if the crop was to be successful as an alternative to sorghum, it must be converted to a short-stature, early-maturing, high-yielding grain type. His breeding work soon attracted attention beyond the borders of Kansas. Beginning in 1978, the project at Hays was linked to international crop improvement programs including the International Crops Institute for Semi-Arid Tropics (ICRISAT) headquartered in Hyderabad, India, and the U. S. Department of State Aid to International Development program entitled International Sorghum and Millet Cooperative Research (INTSORMIL). That cooperation resulted in exchange of several thousand varieties, germplasm lines, and elite inbred selections of pearl millet.



William Stegmeier

Developing hybrids of crops such as sorghum and pearl millet, requires parental lines that are male sterile (that is, plants that will not self pollinate), thus assuring fertilization from a separate male line. Stegmeier's breeding program at Hays developed two male sterile lines from a West Africa source. These lines have been highly successful as parental stock and were recommended by the Indian National Program for general use as seed parents. The new inbred lines imparted early maturity, larger seed size, and higher grain yields than had been available previously. Early maturity is particularly important in areas of India and Africa, where short season crops are desirable because of the need to produce two or even three crops during a given year. Reliable reports from India indicate that by 1997, at least half of all of the pearl millet hybrids marketed by India's private growers were produced using these female parents. This means that more than five million acres of pearl millet grown in India in a given year are hybrids derived from Stegmeier's program. Perhaps even more importantly, these inbred lines are used widely in breeding programs in India and elsewhere. This will lead, in turn, to further improvement in this significant feed and food crop. Because of his work with the international sector, Stegmeier was recog-

nized as part of a team that received a distinguished scientific achievement award sponsored by ICRISAT. Pearl millet is not yet grown widely in Kansas and perhaps never will be, even though extensive yield trials showed that the new hybrids can compete successfully with grain sorghum. Feeding trials with beef cattle indicated that the grain and forage could be used in place of sorghum grain and forage. But the impact of the research program at Hays has been felt worldwide and is particularly meaningful in areas that are often short of food to feed their population.

In addition to the research with pearl millet and sunflower, Stegmeier did a limited amount of work with canola, amaranth, and other possible alternative crops. Of these, probably only canola may find a niche in Kansas agriculture and then only after continued breeding and varietal improvement has taken place. Stegmeier's untimely death from cancer in 1998 brought closure to one of the important crop-improvement programs in the world.

ENTOMOLOGY

Alfalfa flowers must be "tripped" to produce seed. The need to determine the roles of insects and various mechanical devices in alfalfa flower tripping intensified in 1948. Some cooperative work with Agricultural Engineering to study mechanical tripping was underway. The first full-time entomologist arrived at the Station in 1948 to determine the role of both beneficial and harmful insects. This established a unique arrangement between the Station and the Department of Entomology, Kansas State University, in that the entomologist at Hays was a member of the Department staff at Manhattan. This arrangement has continued to the present time and is the only case where a full-time scientist located at Hays is not a Station faculty member.

Dr. Woodrow W. Franklin, who had been working on alfalfa insects in Manhattan, filled the position. He immediately began research to determine the role of wild bees as



Reducing grasshopper infestation with a "hopper dozer." Modern insecticides and insect-resistant crops have reduced losses due to insects.